

## Exercise -I

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Q1. Which of the following statements is true for sets -

- (A) a collection of well defined objects      (B) a collection of objects  
(C) a collection of well defined objects which are distinct and distinguishable  
(D) All of the above

Ans. C

Sol. -

Q2. The set  $\{x : x \in \mathbb{N}, x \text{ is prime and } 3 < x < 5\}$  is-

- (A)  $\{4\}$       (B)  $\{3, 5\}$       (C) Void      (D) Non – Void

Ans. C

Sol. -

Q3.  $A = \{a, e, i, o, u\}$  and  $B = \{i, o\}$  then the true statement is -

- (A)  $A \subset B$       (B)  $B \subset A$       (C)  $A = B$       (D) A is equivalent B

Ans. C

Sol. -

Q4. A set is defined as  $A = \{x : x \text{ is irrational and } 0.1 < x < 0.101\}$  then :

- (A) A is null set      (B) A is finite set      (C) A is infinite set      (D) none

Ans. C

Sol. -

Q5. Which of the following is a singleton Set -

- (A)  $\phi$       (B)  $\{\}$       (C)  $\{2, 3\}$       (D)  $\{\phi\}$

Ans. D

Sol. -

Q6. If  $A = \{\phi, \{\phi\}\}$ , then the power set of A is -

- (A) A      (B)  $\{\phi, \{\phi\}, A\}$       (C)  $\{\phi, \{\phi\}, \{\{\phi\}\}, A\}$       (D) none

Ans. C

Sol. -

Q7. If  $A = \{x | x^2 = 4\}$  and  $B = \{x | x^2 - 5x + 6 = 0\}$  then  $A \cup B$

- (A)  $\{2, 3\}$  (B)  $\{-2, 3\}$  (C)  $\{2, -3\}$  (D)  $\{-2, 2, 3\}$

Ans. D

Sol. -

Q8. Given the sets  $A = \{1, 2, 3\}$ ,  $B = \{3, 4\}$ ,  $C = \{4, 5, 6\}$ , then  $A \cup (B \cap C)$  is -

- (A)  $\{3\}$  (B)  $\{1, 2, 3, 4\}$  (C)  $\{1, 2, 4, 5\}$  (D)  $\{1, 2, 3, 4, 5, 6\}$

Ans. B

Sol. -

Q9. If  $N_a = \{a n : n \in \mathbb{N}\}$ , then  $N_6 \cap N_8 =$

- (A)  $N_6$  (B)  $N_8$  (C)  $N_{24}$  (D)  $N_{44}$

Ans. C

Sol. -

Q10. Which of the following is the empty set ?

- (A)  $\{x : x \in \mathbb{R} \text{ and } x^2 + x + 1 = 0\}$  (B)  $\{x : x \in \mathbb{R} \text{ and } x^2 - x + 1 = 0\}$   
(C)  $\{x : x \in \mathbb{R} \text{ and } x^2 + 2x + 1 \leq 0\}$  (D)  $\{x : x \in \mathbb{R} \text{ and } x^2 - 2x + 1 \geq 0\}$

Ans. A, B

Sol. -

Q11. Two finite sets have  $m$  and  $n$  elements respectively. The total number of subsets of first set is 56 more than the total number of subsets of the second set. The values of  $m$  and  $n$  respectively are -

- (A) 7, 6 (B) 6, 3 (C) 5, 1 (D) 8, 7

Ans. B

Sol. -

Q12. If  $A = \{x | x/2 \in \mathbb{Z}, 0 \leq x \leq 10\}$ ,  $B = \{x | x \text{ is one digit prime}\}$   
 $C = \{x | x/3 \in \mathbb{N}, x \leq 12\}$ , Then  $A \cap (B \cup C)$  is equal to -

- (A)  $\{2, 6\}$  (B)  $\{3, 6, 12\}$  (C)  $\{2, 6, 12\}$  (D)  $\{6, 8\}$

Ans. A

Sol. -

Q13. If  $n(A) = 10$ ,  $n(B) = 15$  and  $n(A \cup B) = x$ , then -

- (A)  $15 < x < 25$     (B)  $15 \leq x \leq 25$     (C)  $5 \leq x \leq 15$     (D) None of these

Ans. B

Sol. -

- Q14. Among 1000 families of a city, 40% read newspaper A, 20% read newspaper B, 10% read newspaper C, 5% read both A and B, 3% read both B and C, 4% read A and C and 2% read all three newspapers. The number of families which read only newspaper A is -

- (A) 140    (B) 290    (C) 330    (D) 340

Ans. C

Sol.  $n(A) = 40\% \text{ of } 1000 = 400$

$$n(B) = 20\% \text{ of } 1000 = 200$$

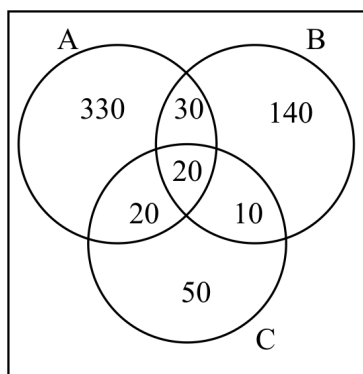
$$n(C) = 10\% \text{ of } 1000 = 100$$

$$n(A \cap B) = 5\% \text{ of } 1000 = 50$$

$$n(B \cap C) = 3\% \text{ of } 1000 = 30$$

$$n(A \cap C) = 4\% \text{ of } 1000 = 40$$

$$n(A \cap B \cap C) = 2\% \text{ of } 1000 = 20$$



The number of families which read only news paper A = 330

- Q15. If for three disjoint sets A, B, C;  $n(A) = 10$ ,  $n(B) = 6$  and  $n(C) = 5$ , then  $n(A \cup B \cup C)$  is equal to -

- (A) 21    (B) 11    (C) 1    (D) 9

Ans. A

Sol. -

- Q16. If A and B are disjoint, then  $n(A \cup B)$  is equal to-

- (A)  $n(A)$     (B)  $n(B)$     (C)  $n(A) + n(B)$     (D)  $n(A).n(B)$

Ans. C

Sol. -

Q17. If  $X$  and  $Y$  are two sets, then  $X \cap (Y \cup X)^c$  equals-

- (A)  $X$  (B)  $Y$  (C)  $\phi$  (D) none

Ans. C

Sol. -

Q18. Let  $n(U) = 700$ ,  $n(A) = 200$ ,  $n(B) = 300$  and  $n(A \cap B) = 100$ , then  $n(A^c \cap B^c)$  is -

- (A) 400 (B) 600 (C) 300 (D) 200

Ans. C

Sol. -

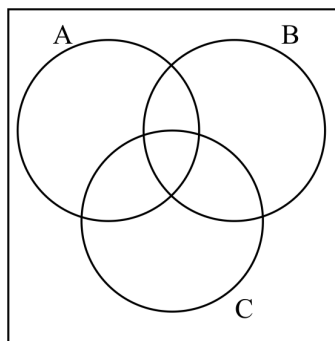
Q19. The set  $(A \cap B^c)^c \cup (B \cap C)$  is equal to -

- (A)  $A \cup B \cup C$  (B)  $A^c \cup B$  (C)  $A^c \cup B^c$  (D) none

Ans. B

Sol.  $(A \cap B^c)^c \cup (B \cap C)$

$(A^c \cup B) \cup (B \cap C)$  {De-morgan's law}



From Venn diagram, clearly see that

$$(A^c \cup B) \cup (B \cap C) = (A^c \cup B)$$

Q20. Sets  $A$  and  $B$  have 3 and 6 elements respectively. What can be the minimum number of elements in  $A \cup B$ ?

- (A) 3 (B) 6 (C) 9 (D) 18

Ans. B

Sol. -

Q21. In a class of 100 students, 55 students have passed in Mathematics and 67 students have passed in physics, no student fails. Then the number of students who have passed in Physics only is-

- (A) 22      (B) 33      (C) 10      (D) 45

Ans. D

Sol. -

Q22. Let  $X = \{1,2,3,4,5,6\}$  be an universal set. Sets A,B,C in the universal set X be defined by

$A = \{1,2,3\}, B = \{2,4,5\}$  &  $C = \{3,4,5,6\}$ . Then-

- (A)  $A-B = \{4,5\}$       (B)  $(A-B) \cup (B-A) = \{1,3,4,5\}$       (C)  $(A-B)-C = \{1\}$   
(D)  $A \cap C' = \{1,2\}$

Ans. D, C, B

Sol. -

Q23. If A, B and C are any three sets, then  $A \times (B \cup C)$  is -

- (A)  $(A \times B) \cup (A \times C)$       (B)  $(A \cup B) \times (A \cup C)$       (C)  $(A \times B) \cap (A \times C)$   
(D) none of these

Ans. A

Sol.  $A \times (B \cup C) = (A \times B) \cup (A \times C)$

Q24.

Ans. -

Sol. -

Q25. Let  $A = \{a, b, c, d\}, B = \{b, c, d, e\}$ . Then  $n[(A \times B) \cap (B \times A)]$  is equal to -

- (A) 3      (B) 6      (C) 9      (D) none

Ans. C

Sol. -

Q26. In the set  $A = \{1, 2, 3, 4, 5\}$ , a relation R is defined by  $R = \{(x, y) \mid x, y \in A \text{ and } x < y\}$ . Then R is -

- (A) Reflexive      (B) Symmetric      (C) Transitive      (D) None of these

Ans. C

Sol. -

- Q27. Let  $R$  be a relation on the set  $N$  of natural numbers defined by  $nRm \Leftrightarrow n$  is a factor of  $m$  (i.e.  $n \mid m$ ). Then  $R$  is -
- (A) Reflexive and symmetric    (B) Transitive and symmetric    (C) Equivalence  
(D) Reflexive, transitive but not symmetric

Ans. D

Sol. -

- Q28. If  $R$  is a relation from a finite set  $A$  having  $m$  elements to a finite set  $B$  having  $n$  elements, then the number of relations from  $A$  to  $B$  is-

(A)  $2^{mn}$     (B)  $2^{mn-1}$     (C)  $2mn$     (D)  $m^n$

Ans. A

Sol.  $n(A) = m$

$n(B) = n$

$n(A \times B) = mn$

no. of subsets of  $(A \times B) = 2^{mn}$

Maximum number of relations from  $A$  to  $B$  = Total no. of subsets of  $(A \times B) = 2^{mn}$

- Q29. Let  $L$  denote the set of all straight lines in a plane. Let a relation  $R$  be defined by  $\alpha R \beta \Leftrightarrow \alpha \perp \beta, \alpha, \beta \in L$ . Then  $R$  is -

(A) Reflexive    (B) Symmetric    (C) Transitive    (D) None of these

Ans. B

Sol.  $L$  denote the set of all straight lines in a plane.

Relation  $R$  be defined by  $R\beta \Leftrightarrow \alpha \perp \beta, \alpha, \beta \in I$

For reflexive,  $\alpha \perp \alpha$  No because a line can't be perpendicular to itself.

For symmetric,  $\alpha \perp \beta \Rightarrow \beta \perp \alpha$ , ycs

for transitive;  $\alpha \perp \beta$  and  $\beta \perp \gamma \Rightarrow$ , No it will be parallel

- Q30. Two points  $A$  and  $B$  in a plane are related if  $OA = OB$ , where  $O$  is a fixed point. This relation is -

(A) Reflexive but not symmetric    (B) Symmetric but not transitive  
(C) An equivalence relation    (D) None of these

Ans. C

Sol. -

- Q31. The relation  $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 3), (1, 3)\}$  on the set  $A = \{1, 2, 3\}$  is -  
(A) Reflexive but not symmetric (B) Reflexive but not transitive  
(C) Symmetric and transitive (D) Neither symmetric nor transitive

Ans. A

Sol. -

- Q32. Let  $A$  be the set of all children in the world and  $R$  be a relation in  $A$  defined by  $x R y$  if  $x$  and  $y$  have same sex. Then  $R$  is –  
(A) Not reflexive (B) Not symmetric (C) Not transitive  
(D) An equivalence relation

Ans. D

Sol. -

- Q33. Let  $A = \{2, 3, 4, 5\}$  and let  $R = \{(2, 2), (3, 3), (4, 4), (5, 5), (2, 3), (3, 2), (3, 5), (5, 3)\}$  be a relation on  $A$ . Then  $R$  is -  
(A) Reflexive and transitive (B) Reflexive and symmetric  
(C) An equivalence relation (D) None of these

Ans. B

Sol. -

- Q34. Let  $L$  be the set of all straight lines in the  $xy$ -plane. Two lines  $\ell_1$  and  $\ell_2$  are said to be related by the relation  $R$  if  $\ell_1$  is parallel to  $\ell_2$ . Then the relation  $R$  is -  
(A) Reflexive (B) Symmetric (C) Transitive (D) Equivalence

Ans. D

Sol. -

- Q35. Given the relation  $R = \{(2, 3), (3, 4)\}$  on the set  $\{2, 3, 4\}$ . The number of minimum number of ordered pairs to be added to  $R$  so that  $R$  is reflexive and symmetric –  
(A) 4 (B) 5 (C) 6 (D) 7

Ans. B

Sol. -

- Q36. The minimum number of elements that must be added to the relation  $R = \{(1, 2), (2, 3)\}$  on the set  $\{1, 2, 3\}$ , so that it is equivalence is-  
(A) 4 (B) 7 (C) 6 (D) 5

Ans. B

Sol. -

Q37. Let a relation 'R' is define on 'Z' set of integers such that  $a R b \Rightarrow a$  is divisible by  $b$  then 'R' is-

(A) Reflexive (B) Symmetric (C) Transitive (D) Equivalence relation

Ans. C

Sol. -

Q38. Let  $R : \Delta \rightarrow \Delta$ , where  $\Delta$  is set of all triangles such that  $\Delta_1 R \Delta_2 \Rightarrow \Delta_1$  is congruent to  $\Delta_2$  then 'R' is-

(A) Reflexive (B) Symmetric (C) Transitive (D) Equivalence relation

Ans. D

Sol. -

Q39. Let  $R : A \rightarrow A$ , A is set of all children in a family such that  $x R y \Rightarrow x$  is brother of 'y' (where  $x, y \in A$ ), then R is-

(A) Reflexive (B) Symmetric (C) Transitive (D) Equivalence relation

Ans. C

Sol. -

## Exercise -II

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Q1. Let  $A = \{1, 2, 3, 4\}$ , and let  $R = \{(2, 2), (3, 3), (4, 4), (1, 2)\}$  be a relation on  $A$ . Then  $R$  is-

- (A) Reflexive    (B) Symmetric    (C) Transitive    (D) None of these

Ans. C

Sol. -

Q2. The void relation on a set  $A$  is -

- (A) Reflexive    (B) Symmetric and transitive    (C) Reflexive and symmetric  
(D) Reflexive and transitive

Ans. B

Sol. -

Q3. Let  $R$  be a relation on the set  $N$  of natural numbers defined by  $nRm \Leftrightarrow n$  is a factor of  $m$  (i.e.  $n \mid m$ ). Then  $R$  is -

- (A) Reflexive and symmetric    (B) Transitive and symmetric    (C) Equivalence  
(D) Reflexive, transitive but not symmetric

Ans. D

Sol. -

Q4. For real numbers  $x$  and  $y$ , we write  $x R y \Leftrightarrow x - y + \sqrt{2}$  is an irrational number. Then the relation  $R$  is -

- (A) Reflexive    (B) Symmetric    (C) Transitive    (D) None of these

Ans. A

Sol. -

Q5. Let  $X = \{1, 2, 3, 4, 5\}$  and  $Y = \{1, 3, 5, 7, 9\}$ . Which of the following is/are relations from  $X$  to  $Y$  -

- (A)  $R_1 = \{(x, y) \mid y = 2 + x, x \in X, y \in Y\}$   
(B)  $R_2 = \{(1, 1), (2, 1), (3, 3), (4, 3), (5, 5)\}$   
(C)  $R_3 = \{(1, 1), (1, 3), (3, 5), (3, 7), (5, 7)\}$     (D)  $R_4 = \{(1, 3), (2, 5), (2, 4), (7, 9)\}$

Ans. A, B, C

Sol. -

- Q6. Let  $R$  be a relation defined in the set of real numbers by  $a R b \Leftrightarrow 1 + ab > 0$ . Then  $R$  is-
- (A) Equivalence relation (B) Transitive (C) Symmetric (D) Anti-symmetric
- Ans. C
- Sol. -
- Q7. Which one of the following relations on  $R$  is equivalence relation -
- (A)  $x R_1 y \Leftrightarrow |x| = |y|$  (B)  $x R_2 y \Leftrightarrow x \neq y$  (C)  $x R_3 y \Leftrightarrow x \mid y$
- (D)  $x R_4 y \Leftrightarrow x < y$
- Ans. A
- Sol. -
- Q8. Let  $R$  be a relation in  $N$  defined by  $R = \{(1+x, 1+x^2) : x \leq 5, x \in N\}$ . Which of the following is false -
- (A)  $R = \{(2, 2), (3, 5), (4, 10), (5, 17), (6, 25)\}$  (B) Domain of  $R = \{2, 3, 4, 5, 6\}$
- (C) Range of  $R = \{2, 5, 10, 17, 26\}$  (D) None of these
- Ans. A
- Sol. -
- Q9. The relation  $R$  defined in  $A = \{1, 2, 3\}$  by  $a R b$  if  $|a^2 - b^2| \leq 5$ . Which of the following is false
- (A)  $R = \{(1, 1), (2, 2), (3, 3), (2, 1), (1, 2), (2, 3), (3, 2)\}$  (B)  $R^{-1} = R$
- (C) Domain of  $R = \{1, 2, 3\}$  (D) Range of  $R = \{5\}$
- Ans. D
- Sol. -
- Q10. The relation  $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 3), (1, 3)\}$  on the set  $A = \{1, 2, 3\}$  is -
- (A) Reflexive but not symmetric (B) Reflexive but not transitive
- (C) Symmetric and transitive (D) Neither symmetric nor transitive
- Ans. A
- Sol. -
- Q11. Let a relation  $R$  in the set  $N$  of natural numbers be defined as  $(x, y) \in R$  if and only if  $x^2 - 4xy + 3y^2 = 0$  for all  $x, y \in N$ . The relation  $R$  is -

(A) Reflexive (B) Symmetric (C) Transitive (D) An equivalence relation

Ans. A

Sol. -

Q12. Let  $A = \{2, 3, 4, 5\}$  and let  $R = \{(2, 2), (3, 3), (4, 4), (5, 5), (2, 3), (3, 2), (3, 5), (5, 3)\}$  be a relation in A. Then R is -

(A) Reflexive and transitive (B) Reflexive and symmetric

(C) Reflexive and antisymmetric (D) None of these

Ans. B

Sol. -

Q13. If  $A = \{2, 3\}$  and  $B = \{1, 2\}$ , then  $A \times B$  is equal to -

(A)  $\{(2, 1), (2, 2), (3, 1), (3, 2)\}$  (B)  $\{(1, 2), (1, 3), (2, 2), (2, 3)\}$

(C)  $\{(2, 1), (3, 2)\}$  (D)  $\{(1, 2), (2, 3)\}$

Ans. A

Sol. -

Q14. Let  $R = \{(3, 3), (6, 6), (9, 9), (12, 12), (6, 12), (3, 9), (3, 12), (3, 6)\}$  be a relation on the set  $A = \{3, 6, 9, 12\}$ . The relation is -

(A) An equivalence relation (B) Reflexive and symmetric only

(C) Reflexive and transitive only (D) Reflexive only

Ans. C

Sol. -

Q15. Let  $R = \{(1, 3), (4, 2), (2, 4), (2, 3), (3, 1)\}$  be a relation on the set  $A = \{1, 2, 3, 4\}$ . The relation R is -

(A) Reflexive (B) Transitive (C) Not symmetric (D) A function

Ans. C

Sol. -

Q16. Let  $N$  denote the set of all natural numbers and  $R$  be the relation on  $N \times N$  defined by  $(a, b) R (c, d)$  if  $ad(b + c) = bc(a + d)$ , then R is-

(A) Symmetric only (B) Reflexive only (C) Transitive only

(D) An equivalence relation

Ans. D

Sol. -

Q17. If  $A = \{1, 2, 3\}$ ,  $B = \{1, 4, 6, 9\}$  and  $R$  is a relation from  $A$  to  $B$  defined by ' $x$  is greater than  $y$ '. The range of  $R$  is -

(A)  $\{1, 4, 6, 9\}$  (B)  $\{4, 6, 9\}$  (C)  $\{1\}$  (D) None of these

Ans. C

Sol. -

## Exercise -III

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Q1.

Ans. -

Sol. -

Q2. Let  $R = \{(3, 3), (6, 6), (9, 9), (12, 12), (6, 12), (3, 9), (3, 12), (3, 6)\}$ , be relation on the set  $A = \{3, 6, 9, 12\}$ . The relation is -

- (A) reflexive and transitive only    (B) reflexive only    (C) an equivalence relation  
(D) reflexive and symmetric only

Ans. A

Sol. -

Q3. Let  $W$  denote the words in the English dictionary. Define the relation  $R$  by : $R = \{(x, y) \in W \times W \mid \text{the words } x \text{ and } y \text{ have at least one letter in common}\}$ . Then  $R$  is –

- (A) reflexive, symmetric and not transitive    (B) reflexive, symmetric and transitive  
(C) reflexive, not symmetric and transitive  
(D) not reflexive, symmetric and transitive

Ans. A

Sol. -

Q4.

Ans. -

Sol. -

Q5. If  $A$ ,  $B$  and  $C$  are three sets such that  $A \cap B = A \cup C$  and  $A \cup B = A \cup C$ , then -

- (A)  $A = B$     (B)  $A = C$     (C)  $B = C$     (D)  $A \cap B = \phi$

Ans. C

Sol. -

Q6. Consider the following relations :-

$R = \{(x, y) \mid x, y \text{ are real numbers and } x = wy \text{ for some rational number } w\}$ ;

$S = \{(\frac{m}{n}, \frac{p}{q}) \mid m, n, p \text{ and } q \text{ are integers such that } n, q \neq 0 \text{ and } qm = pn.\}$ , Then

- (A)  $R$  is an equivalence relation but  $S$  is not an equivalence relation  
(B) Neither  $R$  nor  $S$  is an equivalence relation

(C) S is an equivalence relation but R is not an equivalence relation

(D) R and S both are equivalence relations

Ans. C

Sol. -

Q7. Let R be the set of real numbers.

Statement-1:  $A = \{(x, y) \in \mathbb{R} \times \mathbb{R} : y - x \text{ is an integer}\}$  is an equivalence relation on R.

Statement :2:  $B = \{(x, y) \in \mathbb{R} \times \mathbb{R} : x = \alpha y \text{ for some rational number } \alpha\}$  is an equivalence relation on R.

(A) Statement-1 is true, Statement-2 is false.

(B) Statement-1 is false, Statement-2 is true.

(C) Statement-1 is true, Statement-2 is true; Statement-2 is a correct explanation for Statement-1.

(D) Statement-1 is true, Statement-2 is true; Statement-2 is not a correct explanation for Statement-1.

Ans. 5f3e4078d90daa8b97bb4cb6

Sol. -

Q8. Consider the following relation R on the set of real square matrices of order 3.

$R = \{(A, B) \mid A = P^{-1}BP \text{ for some invertible matrix } P\}$ .

Statement-1: R is an equivalence relation.

Statement-2: For any two invertible  $3 \times 3$  matrices M and N,  $(MN)^{-1} = N^{-1}M^{-1}$

(A) Statement-1 is false, statement-2 is true.

(B) Statement-1 is true, statement-2 is true; Statement-2 is correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is true; Statement-2 is not a correct explanation for statement-1.

(D) Statement-1 is true, statement-2 is false.

Ans. 5f3e419f0f9f218bad35bc3d

Sol. -

Q9. Let  $X = \{1, 2, 3, 4, 5\}$ . The number of different ordered pairs (Y, Z) that can be formed such that  $Y \subseteq X$ ,  $Z \subseteq X$  and  $Y \cap Z$  is empty, is

(A)  $5^3$  (B)  $5^2$  (C)  $3^5$  (D)  $2^5$

Ans. C

Sol. -

Q10. Let A and B be two sets containing 2 elements and 4 elements respectively. The number of subsets of  $A \times B$  having 3 or more elements is

(A) 256 (B) 220 (C) 219 (D) 211

Ans. C

Sol. -

Q11. Let  $S = \{x \in \mathbb{R} : x \geq 0 \text{ and } 2\sqrt{x} - 3 + \sqrt{x}(\sqrt{x} - 6) + 6 = 0\}$ . Then S :

(A) contains exactly two elements (B) contains exactly four elements  
(C) is an empty set (D) contains exactly one element

Ans. A

Sol. -

Q12. Let  $A = \{n \in \mathbb{N} : n \text{ is a 3-digit number}\}$

$B = \{9k + 2 : k \in \mathbb{N}\}$

and  $C = \{9k + l : k \in \mathbb{N}\}$  for some  $l$  ( $0 < l < 9$ )

If the sum of all the elements of the set  $A \cap (B \cup C)$  is  $274 \times 400$ , then  $l$  is equal to

\_\_\_\_\_.

Ans. 5

Sol. If  $B \cup C = B$

$$\sum_{K=11}^{110} (9k + 2) = 9 \left( \sum_{K=1}^{110} K - \sum_{K=1}^{10} K \right) + 200 \\ = 54650 \neq 274 \times 400$$

If  $B \cup C \neq B$

$$\sum_{K=11}^{110} (9k + 2) + \sum_{K=11}^{110} (9k + l) \times 400 \\ \ell = 5$$

Q13. Let  $R = \{(P, Q) \mid P \text{ and } Q \text{ are at the same distance from the origin}\}$  be a relation, then the equivalence class of  $(1, -1)$  is the set :

(A)  $S = \{(x, y) \mid x^2 + y^2 = \sqrt{2}\}$  (B)  $S = \{(x, y) \mid x^2 + y^2 = 2\}$  (C)  $S = \{(x, y) \mid x^2 + y^2 = 4\}$   
(D)  $S = \{(x, y) \mid x^2 + y^2 = 1\}$

Ans. B

Sol.  $R(1, -1) P(h, k), O(0, 0)$   
 $OR = OP$

locus of P is  $h^2 + k^2 = 2$   
 $x^2 + y^2 = 2$

Q14. Define a relation R over a class of  $n \times n$  real matrices A and B as "ARB iff there exists a non-singular matrix P such that  $PAP^{-1} = B$ " Then which of the following is true ?

- (A) R is reflexive, transitive but not symmetric      (B) R is an equivalence relation  
 (C) R is symmetric, transitive but not reflexive  
 (D) R is reflexive, symmetric but not transitive

Ans. B

Sol. For reflexive

$$ARA \Rightarrow PAP^{-1} = A \dots (1) \text{ must be true}$$

For  $P = I$ , eq. (1) is true so 'R' is reflexive.

For symmetric

$$ARB \Leftrightarrow PAP^{-1} = B \dots (1) \text{ is true}$$

For BRA iff  $PBP^{-1} = A \dots (2) \text{ must be true}$

$$\therefore PAP^{-1} = B$$

$$P^{-1}PA P^{-1} = P^{-1}B$$

$$IAP^{-1}P = P^{-1}BP$$

$$A = P^{-1}BP \dots (3)$$

From (2) and (3)  $PBP^{-1} = P^{-1}BP$  can be true for some  $P = P^{-1}$

$$\Rightarrow P^2 = I \text{ (det (P) } \neq 0)$$

So, R is symmetric

For transitive

$$ARB \Leftrightarrow PAP^{-1} = B \dots \text{ is true}$$

$$BRC \Leftrightarrow PBP^{-1} = C \dots \text{ is true}$$

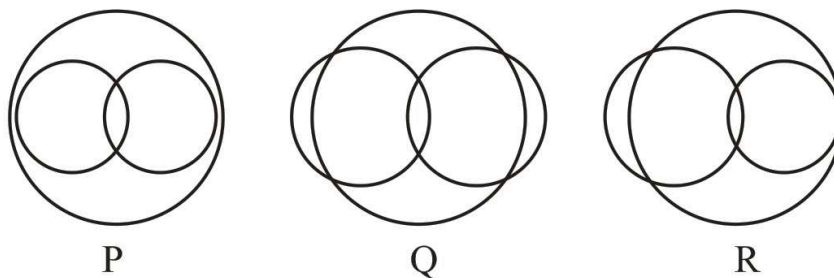
$$\text{Now } PPA P^{-1} P^{-1} = C$$

$$P^2A (P^2)^{-1} = C \Leftrightarrow ARC$$

So R is transitive relation

$\Rightarrow$  Hence, R is equivalence

Q15. In a school, there are three types of games to be played. Some of the students play two types of games, but none play all the three games. Which Venn diagrams can justify the above statement ?



- (A) P and Q    (B) Q and R    (C) None of these    (D) P and R

Ans. C

Sol. None of P, Q, R represent the exact condition given in the problem. So option (C) is right answer

Q16. Let  $A = \{2, 3, 4, 5, \dots, 30\}$  and ' $\simeq$ ' be an equivalence relation on  $A \times A$ , defined by (a, b) (c, d), if and only if  $ad = bc$ . Then the number of ordered pairs which satisfy this equivalence relation with ordered pair (4, 3) is equal to :

- (A) 5    (B) 8    (C) 6    (D) 7

Ans. D

Sol.  $(a, b) \simeq (c, d) \Leftrightarrow \frac{a}{b} = \frac{c}{d}$

$$\frac{4}{3} = \frac{8}{6} = \frac{12}{9} = \frac{16}{12} = \frac{20}{15} = \frac{24}{18} = \frac{28}{21}$$

Ordered pairs (4, 3), (8, 6), (12, 9), (16, 12), (20, 15), (24, 18), (28, 21)

Total = 7

Q17. The number of elements in the set  $\{x \in \mathbb{R} : (|x| - 3) |x + 4| = 6\}$  is equal to :

- (A) 2    (B) 1    (C) 3    (D) 4

Ans. A

Sol.  $(|x| - 3) |x + 4| = 6$

**Case - I**  $x \geq 0$

$$(x - 3)(x + 4) = 6$$

$$x^2 + x - 12 = 6$$

$$x^2 + x - 18 = 0$$

$$x = \frac{-1 \pm \sqrt{73}}{2} \Rightarrow x = \frac{1 + \sqrt{73}}{2} > 0$$

**Case - II**  $-4 \leq x < 0$

$$(-x - 3)(x + 4) = 6$$

$$x^2 + 7x + 12 + 6 = 0$$

$$x^2 + 7x + 18 = 0 \quad D < 0$$

**Case - III**  $x < -4$

$$(-x-3)(-x-4) = 6$$

$$x^2 + 7x + 12 = 6$$

$$x^2 + 7x + 6 = 0$$

$$(x+1)(x+6) = 0$$

$$x = -1, x = -6 \Rightarrow x = -6$$

Q18. Let  $N$  be the set of natural numbers and a relation  $R$  on  $N$  be defined by

$R = \{(x, y) \in N \times N : x^3 - 3x^2y - xy^2 + 3y^3 = 0\}$ . Then the relation  $R$  is :

(A) reflexive but neither symmetric nor transitive

(B) symmetric but neither reflexive nor transitive (C) an equivalence relation

(D) reflexive and symmetric, but not transitive

Ans. A

Sol.

$$(x, y) \in N \times N$$

$$x^3 - 3x^2y - xy^2 + 3y^3 = 0$$

$$\Rightarrow x(x^2 - y^2) - 3y(x^2 - y^2) = 0$$

$$\Rightarrow (x - 3y)(x^2 - y^2) = 0$$

$$\Rightarrow (x - y)(x + y)(x - 3y) = 0$$

$$\therefore x = y \text{ or } x = -y \text{ or } x = 3y$$

It is a reflexive relation Since  $xRx$

$$\therefore x = 3y$$

Consider  $(3, 1) \in R$

But  $(1, 3) \notin R$

$\therefore$  Hence it is not Symmetric

For transitive Consider  $(9, 3) \in R$  and  $(3, 1) \in R$

but  $(9, 1) \notin R$

$\therefore$  It is not transitive.

Q19. Let  $A = \{n \in N \mid n^2 + 10,000\}$ ,  $B = \{3k + 1 \mid k \in N\}$  and  $C = \{2k \mid k \in N\}$ , then the sum of all the elements of the set  $A \cap (B - C)$  is equal to \_\_\_\_.

Ans. 832

Sol.  $A = \{n \in N \mid n^2 \leq n + 1000\}$

$$B = \{3k + 1 \mid k \in N\}$$

$$C = \{2k \mid k \in N\}$$

$$\therefore \text{For } n^2 \leq n + 10000$$

$$\Rightarrow n(n-1) \leq 10000$$

$\therefore$  Product of two consecutive nature No.  $\leq 10000$

$$\therefore A = \{1, 2, 3, 4, \dots, 100\}$$

$$\therefore B = \{4, 7, 10, \dots, 97, 100, 103, \dots\}$$

$$C = \{2, 4, 6, 8, \dots, 98, 100, 102, \dots\}$$

$$\therefore B - C = \{7, 13, 19, \dots, 91, 97\}$$

$$A \cap (B - C) = \{7, 13, 19, \dots, 91, 97\}$$

No. of elements = 16

Sum of all the elements in  $A \cap (B - C)$

$$= 7 + 13 + 19 + \dots + 91 + 97$$

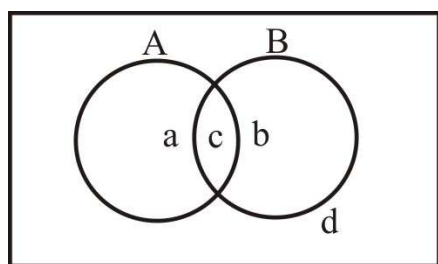
$$= \frac{16}{2} [7 + 97] = 16 \times 52 = 832$$

Q20. Out of all the patients in a hospital 89% are found to be suffering from heart ailment and 98% are suffering from lungs infection. If K% of them are suffering from both ailments, then K can not belong to the set :

- (A)  $\{84, 87, 90, 93\}$     (B)  $\{79, 81, 83, 85\}$     (C)  $\{84, 86, 88, 90\}$   
 (D)  $\{80, 83, 86, 89\}$

Ans. B

Sol.



$A \equiv$  Heart ailment ;  $B \equiv$  lung ailment

$$a + b + c + d = 100 \quad \dots\dots\dots(1)$$

$$a + c = 89 \quad \dots\dots\dots(2)$$

$$b + c = 98 \quad \dots\dots\dots(3)$$

$$a + b + 2c = 187$$

$$a + b + c \leq 100$$

$$\Rightarrow c \geq 87$$

Q21. Which of the following is not correct for relation R on the set of real numbers ?

- (A)  $(x, y) \in R \Leftrightarrow 0 < |x| - |y| \leq 1$  is neither transitive nor symmetric  
 (B)  $(x, y) \in R \Leftrightarrow 0 < |x| - |y| \leq 1$  is reflexive and symmetric

(C)  $(x, y) \in R \Leftrightarrow 0 < |x - y| \leq 1$  is symmetric and transitive

(D)  $(x, y) \in R \Leftrightarrow 0 < |x| - |y| \leq 1$  is reflexive but not symmetric

Ans. C

Sol.  $(x, y) \in R \Leftrightarrow 0 < |x - y| \leq 1$  is symmetric but not transitive.

Q22. If  $A = \{x \in R : |x - 2| > 1\}$ ,  $B = \{x \in R : \sqrt{x^2 - 3} > 1\}$ ,  $C = \{x \in R : |x - 4| \geq 2\}$  and  $Z$  is the set of all integers, then the number of subsets of the set  $(A \cap B \cap C)^c \cap Z$  is \_\_\_\_\_.

Ans. 256

Sol.  $A : x - 2 < -1 \quad \text{or} \quad x - 2 > 1$   
 $x < 1 \quad \text{or} \quad x > 3$

$x \in (-\infty, 1) \cup (3, \infty)$

$B : \sqrt{x^2 - 3} > 1 \Rightarrow x^2 - 3 > 1$

$\Rightarrow x^2 > 4$

$\Rightarrow x \in (-\infty, -2) \cup (2, \infty)$

$C : x - 4 \leq -2 \quad \text{or} \quad x - 4 \geq 2$

$\Rightarrow x \leq 2 \quad \text{or} \quad x \geq 6$

$\Rightarrow x \in (-\infty, 2] \cup [6, \infty)$

$A \cap B \cap C = (-\infty, -2) \cup [6, \infty)$

$(A \cap B \cap C)^c = [-2, 6)$

$(A \cap B \cap C)^c \cap Z = \{-2, -1, 0, 1, 2, 3, 4, 5\}$

Number of subsets  $= 2^8 = 256$